

Karan Shah

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Education

Stony Brook University

Stony Brook, NY

MS IN COMPUTER SCIENCE | GPA : 3.64/4

Aug. 2019 - **Exp. Dec. 2020**

Coursework: Big Data Analytics, Theory of Database Systems, Machine Learning, Computer Vision, Data Science, Probability and Statistics

Gujarat Technological University

Ahmedabad, India

B.E. IN COMPUTER ENGINEERING

Aug. 2014 - May 2018

Coursework : Data Structures, Algorithms, Software Engineering, Object Oriented Programming, Distributed Operating System

Skills

IDEs Visual Studio, NetBeans, Jupyter Notebook, Eclipse, IntelliJ IDEA, Vim, Sublime

Programming JAVA, Python, Javascript, C/C++

Tools/Technologies AWS, Linux, MapReduce, Spark, Hadoop, Docker, Jenkins, Kubernetes, Git, SQL, NoSQL, Soap, Matlab

Frameworks/Libraries Node.js, AJAX, Django, Flask, React.js, Electron.js, Scikit-Learn, Numpy, Pandas, PyTorch, Tensorflow, Keras, OpenCV

Work Experience

Stony Brook Medicine

Stony Brook, NY

GRADUATE STUDENT RESEARCHER

January 2020 - Current

- Working on a project "RadSearch" under supervision of **Prof. I.V. Ramakrishnan**, a Search Engine used regularly by the doctors at the Radiology Department of the Stony Brook Hospital to find the most relevant information about patients
- Created a distributed, multitenant-capable full-text search engine with an HTTP web interface and schema-free JSON documents using **Elastic Search** to handle user queries and **Java Servlets, Python, Javascript** to handle the backend

Simform

Ahmedabad, India

SOFTWARE ENGINEER

June 2018 - April 2019

- Designed and automated **ETL** jobs to process a huge realtime data using AWS services (Glue, Lambda, S3) and Spark
- Developed an end to end **Video Streaming system**; able to upload, scan and host the content on the cloud, and stream on cross-platform desktop application over globally separated clients; using **AWS services (Lambda, S3, Cloudfront, MQTT), NodeJS, and Python**
- Built a software system to semi-automate Company's marketing work with Chrome Extension and **REST API** Web backend using **Python and Node.js**, efficient in reducing ~ **40 %** of Marketing team's manual work

ISRO - Indian Space Research Organization

Ahmedabad, India

RESEARCH INTERN, MACHINE LEARNING

July 2017 - May 2018

- Designed generalized **CNN** architecture to handle multimodal and/or multispectral imagery along with adaptability to varying application requirements; trained on Potsdam benchmark dataset producing State of the art results to **Semantic Segmentation**
- Built software tools in **Python** for data preparation/processing of geographic remote sensing data, visualization, real-time training performance evaluation, post-processing and output generation as a semi-automation approach
- Manipulated raster GIS data as a part of the pre-processing task using Python and QGIS application

Projects

Child Mortality Analysis, Under Supervision of Prof. Andrew Schwartz (Python, GCP, Spark)

April 2020 - May 2020

- Analysed secondary factors contributing to child mortality through Big Data Techniques
- Implemented **Linear Regression and Similarity Search** running on women lifestyle real data distributed on Hadoop clusters

Retail Sales Data Analysis, Under Supervision of Prof. Steven Skiena (Python, Pandas)

Oct. 2019 - Dec. 2019

- Provided insights to a local chain retail stores to help the company with their business by working on their market basket dataset
- Analysed data to find reasons for drop in sales with additional help of external datasets
- Implemented **Word2Vec and LSTM model with k-means clustering** to improve Shelf Space Management and Inventory management

Video Action Classification and Recognition (Python, PyTorch, Matlab)

Nov. 2019 - Dec. 2019

- Trained CNN for human action recognition task on the UCF101 data
- Achieved **86%** (inclass 3rd rank) on image test data, and **84%** (inclass 4th rank) on video frames test data
- Trained LSTM in RNN to actions classification on data collected by Kinect v2
- Used Transfer Learning to compute features for 60000 video frames with limited compute resources.

Company's Internal Communication System (Java, Javascript)

Sept. 2018 - Dec. 2018

- Created a web-based software for employees of a Software Company to send/receive messages, share files in person, groups and broadcast channels with various administrative permission roles
- Utilised **SQLite** to store data of each user with **Java** Servlets to query the database and applied Software Engineering approaches

Desktop Application to Synchronize Local Storage with Cloud (AWS, Node.js, Electron.js)

March 2017 - Apr. 2017

- Built cross-platform application leveraging the ElectronJS framework
- Used **Node.js and AWS services** to fully synchronize local storage and AWS S3 storage

Publication

Band-wise Independent Pansharpening Using Neural Networks with Shared Weights

PANKAJ BODANI, KARAN SHAH, SHASHIKANT SHARMA

- Submitted to Geocarto International Journal